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# Probabilistic approach for optimal portfolio selection using a hybrid Monte Carlo simulation and Markowitz model



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Exceedance probability;
robustness

**Abstract** In this paper, a probabilistic form of the portfolio selection problem is established in which the uncertainty of risky assets is considered through a probabilistic optimization problem. To this end, by taking seven portfolios of Shanghai stock as a case study, the mean and standard deviation of daily return values are calculated based on five years of real data. The optimal values corresponding to each random case were then stored as a comprehensive database of system responses. Then, by sorting the resulting optimal values from best to worst, the exceedance probabilities of return and risk rankings were calculated for each portfolio and presented in the form of probabilistic pie charts. The results showed that the portfolio with the highest deterministic rate of return has the highest probability of getting the best return ranking as well. However, since the probability of risk in all cases was calculated, the probability of each portfolio to place in the lower rankings (i.e., ranking 2–7) could also be discussed. Additionally, to check the convergence of the model, the probability values calculated by the Monte Carlo method against the sample size were plotted to ensure the accuracy of the final answer. Eventually, by generating Gaussian random noise and importing it into the model input, probability changes were calculated to assess the robustness of the proposed algorithm.

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## 1. Introduction

The capital market is one of the market places where the holder of financial resources, on the one hand, and organizations, on the other hand, are actively involved in financial activities [1,2]. The task of the capital market is to provide financial resources to those in need, and the share is known as the most important commodity traded in the capital market [3,4]. Most people make some investment decisions in their lives [5], and capital market is one of the most important places to help optimize asset allocation [6,7]. Individuals in their analyses eventually hope to get an appropriate return with low investment risk [8–11], but one of the essential features of the capital market is uncertainty [12,13]. The price fluctuations make the investors worried about the future of their capitals [14,15]. For investors, achieving the high return is a positive parameter, while the volatility is a negative parameter

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[16,17]. The question here is how investors should allocate their existing capitals to different stocks, that is, what fraction of portfolio should be allocated to existing stocks [18,19]. Therefore, the portfolio optimization is of considerable significance in the finance and investment projects [20–22]. The study and analysis of the basic principles of portfolio management can lead to a better financial plan and increase the wealth of investors [23].

The theory of portfolio selection was first put forward by Harry Markowitz, an American economist and Nobel Prize winner [24]. In his paper in 1952, he postulated that investors behave reasonably, always trying to maximize the expected return and reduce the overall risk of the portfolio, taking into account the possible correlation between assets. The theory he proposed in his research later formed the basis for modern portfolio theory (MPT). The Markowitz's concept has been developed by different researchers. In 1963, William Sharpe [25] developed an effective calculation method to analyze available capital and build a portfolio with expected attributes. Later, Robert Merton [26] published an article in which he showed how to analytically calculate the "efficient Pareto frontier". Pogue [27] later extended the model to include "transaction costs" and short selling. In 1992, Black and Litterman [28] proposed a model based on the Markowitz model where no asset with specific expected returns is required to be considered during the analysis. In 1991, Rom and Ferguson [29] extended the model by removing the limitations of the original model, considering the distribution of returns and taking the variance of returns as a measure of investment risk. The work done after this research is often referred to as the "postmodern portfolio theory". Much new research is still being done on the basis of Markowitz's theory, which utilizes the latest modeling and problem-solving approaches. Woodside-Oriakhi et al. [30] introduced a comprehensive model of portfolio optimization and implemented the model with quadratic programming of complex integers. Mittal and Mehlawat [31] proposed a model involving transaction costs and also solved the optimization problem by adopting the genetic algorithm. A more comprehensive review on deterministic portfolio optimization can be found in [32].

To the best of the authors' knowledge, most existing studies on portfolio optimization have used the deterministic methods for the problem formulation [33]. However, the unknown nature of the uncertainty and market volatility involved in the portfolio project is so large that it could not be ignored [34]. In fact, the deterministic modeling is equivalent to assuming 100% certainty about the status of stocks, which is unacceptable in the reality [35]. Hence, an alternative approach is needed to consider the uncertainty and probabilistically address the problem. In this study, an optimal probability algorithm is used for this purpose. In this way, a deterministic optimization algorithm is first implemented for the problem. Then, the input parameters (including the daily data of seven different portfolios of the Shanghai stock market) are modeled as random variables with the known mean and standard deviation. The resulting random variables are then incorporated into the optimization algorithm to establish a systematic approach to the formulation of the probabilistic optimization. The Monte Carlo sampling method is then used to solve the established problem and to present a probabilistic solution for the individual return and risk values [36,37]. Fig. 1 provides an overview of this process. The convergence of the Monte Carlo method is then evaluated to ensure that the number of samples is sufficient and that the algorithm converges to the solution with a reasonable accuracy. Finally, the input data are polluted with Gaussian random noise to assess the robustness of the proposed algorithm to random fluctuations.

## 2. Review of Shanghai stock exchange (SSE)

### 2.1. History of SSE

As mentioned in the introduction, the algorithm proposed in this paper is applied to the SSE. Shanghai is the first city in China that welcomed the stock exchange. SSE opened in the 1860s. In 1891, the Shanghai Share Brokers Association was established in Shanghai. Later, in the 1920s, with the establishment of the Shanghai Security Goods Exchange and the Shanghai Securities Exchange in China, Shanghai became the trading center in the Far East, where the Chinese and foreign investors could trade the stocks.

Since 1980, the China's securities market has grown with the reforms and opening in the country and the development of a socialist market economy. In 1981, the treasury bond issue was resumed. In 1984, the stocks and enterprise bonds were issued in Shanghai and other cities. On November 26, 1990, the SSE was established, and on December 19 of that year, the trading officially started.

### 2.2. Index introduction

SSE is the largest stock exchange in the mainland China run by the China Securities Regulatory Commission as a non-profit organization. It trades stocks, funds, and bonds. To be listed on the Shanghai exchange, a company must have earned profit for at least three years. Most of the exchange in terms of the total market capitalization is comprised of formally state-run companies, including major banks and insurance companies. The exchange uses the A-shares and B-shares. The A-shares are the shares of companies based on mainland China and are quoted in Yuan, but previously available for the mainland citizens. However, recent changes and regulations allow the foreign investors to buy the shares through the tightly regulated system for the qualified foreign investors. The B-shares are quoted in foreign currencies such as the US Dollar and open to domestic and foreign investors.

In this study, the data of seven different portfolio indices of the Shanghai stock market are used as listed in Table 1. A brief introduction of the portfolio indices is provided in the Appendix A.

## 3. Modeling and results

As noted in the introduction, this study attempts to express the optimal portfolio selection in the form of a probabilistic problem, because the deterministic models are generally limited to a particular scenario, even for real-life cases that are associated with a high degree of uncertainty [38–40]. However, one must first start with a deterministic model to calculate the optimal solution of the problem for a particular input vector. Then, the deterministic model can be transformed into a probabilistic model by defining the input parameters in the form of random

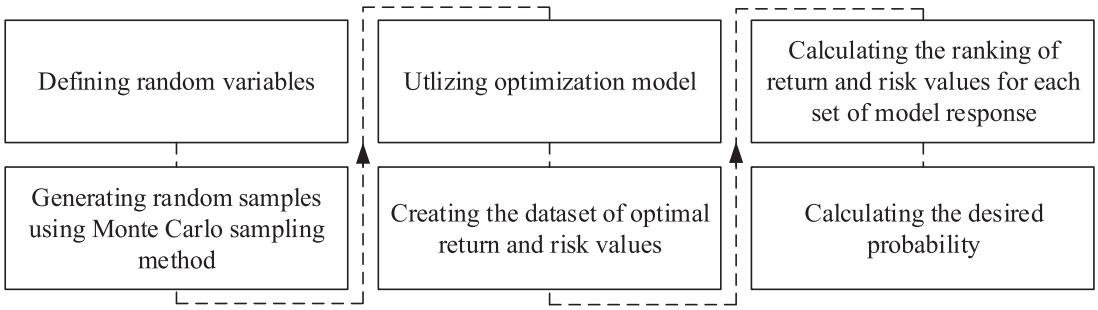



**Fig. 1** The process to establish probabilistic model.

**Table 1** Seven portfolios of Shanghai stock market used in this research.

| No. | Symbol | Associated stock index |
|---|---|---|
| 1 | Portfolio A | SSE Composite Index 000001 |
| 2 | Portfolio B | SSE 180 index 000010 |
| 3 | Portfolio C | SSE 50 Index 000016 |
| 4 | Portfolio D | SSE 380 Index 000009 |
| 5 | Portfolio E | SSE Fund Index 000011 |
| 6 | Portfolio F | T-Bond index |
| 7 | Portfolio G | SHSE-SZSE300 index |

variables [41]. Next, probabilistic analysis methods can be used to solve the recorded problem. Finally, post-processing is required to extract and report the desired response from the randomly generated database. A schematic view of the proposed algorithm is shown in Fig. 2.

### 3.1. Data collection and deterministic portfolio optimization

For this study, the data on the previously-described seven major SSE indices were collected on a daily basis for five years (from 2014 to 2018). First, given the daily close prices of each index, the daily returns were obtained by the following formula:

$$r_i(t) = \frac{p_i(t+1) - p_i(t)}{p_i(t)} = \frac{p_i(t+1)}{p_i(t)} - 1, \tag{1}$$

where $r_i(t)$ represents the return at time $t$, $p_i(t)$ denotes the closing price at time $t$, and $p_i(t+1)$ represents the closing price at time $t+1$. Then, the summation of the daily return values was calculated for each month and considered as monthly return.

As mentioned earlier, the optimization models can provide stocks arrangement with the lowest risk over expected return (or correspondingly highest expected return over risk). For this means, the Markowitz mean–variance model is adopted in this paper as follows:

$$\begin{aligned} Max\{\mu_p\} &= \sum_i w_i \mu_i \\ Min\{\sigma_p^2\} &= \sum_i \sum_j w_i w_j \sigma_{ij} \\ \text{Subject to} &: \begin{cases} \sum_i w_i = 1 \\ w_i \geqslant 0 \end{cases}, \end{aligned} \tag{2}$$

where $\mu_p$ represents the portfolio return, $\sigma_p^2$ denotes the portfolio variance (risk), $\sigma_{ij}$ represents the covariance between the two stocks $i$ and $j$, and $w_i$ is regarded as the share invested in stock $i$. As shown in Eq. (2), the problem is formulated as a multi-objective optimization. The main purpose of this study is to maximize the return and minimize the risk. Therefore, two objective functions were required to define the problem. Of course, we could formulate the problem by only one of these two objective functions, i.e., either maximize return or minimize risk, which would have made the problem more straightforward, but it would not provide a comprehensive answer. For this reason, it was decided to add complexity to the problem, but cover both objectives at the same time. Additionally, it should be noted that different methodologies, including intelligent algorithms, can be used to solve the established portfolio problem. However, intelligent methods are usually implemented through a population-based approach. That is, the initial population should be generated first, and then updated in each increment to gradually approach the optimal solution. The combination of such methods and Monte Carlo analysis (which itself is based on random sample generation) will impose a high computational cost, and consequently, challenge the ease of analysis. Therefore, the authors decided to use the Markowitz method for deterministic portfolio selection. In the following, the Markowitz mean–variance optimization model was implemented in MATLAB. Then, the implemented algorithm was applied to the data obtained from seven different portfolio indices of the Shanghai stock market, and the desired results were consequently obtained. In this model, the summation of daily return in each month is used to estimate the expected return, and the variance is utilized to determine the risk. One hundred strategies were adopted to run the code and draw the efficient Pareto frontier for the seven stock indices using the monthly return data of each index (each scenario means an optimal response). The results are shown in Fig. 3.

In this graph, the horizontal axis represents the risk amount, and the vertical axis denotes the return of different combinations of seven stock indexes in this study. As can be seen from the figure, the SSE180 index has the highest return but also the highest risk among all indexes.

According to Fig. 3, SSE180 Index, SSE50 Index, Fund Index, SSE300 Index, SSE Composite Index, SSE380 Index, and T-Bond Index have the highest returns, respectively. Additionally, the risk indices ordered from low to high are: T-Bond Index, Fund Index, SSE Composite Index, SHSE300 Index, SSE380 Index, SSE50 Index, and SSE180 Index. The efficient frontier graph represents 100 different combinations of the

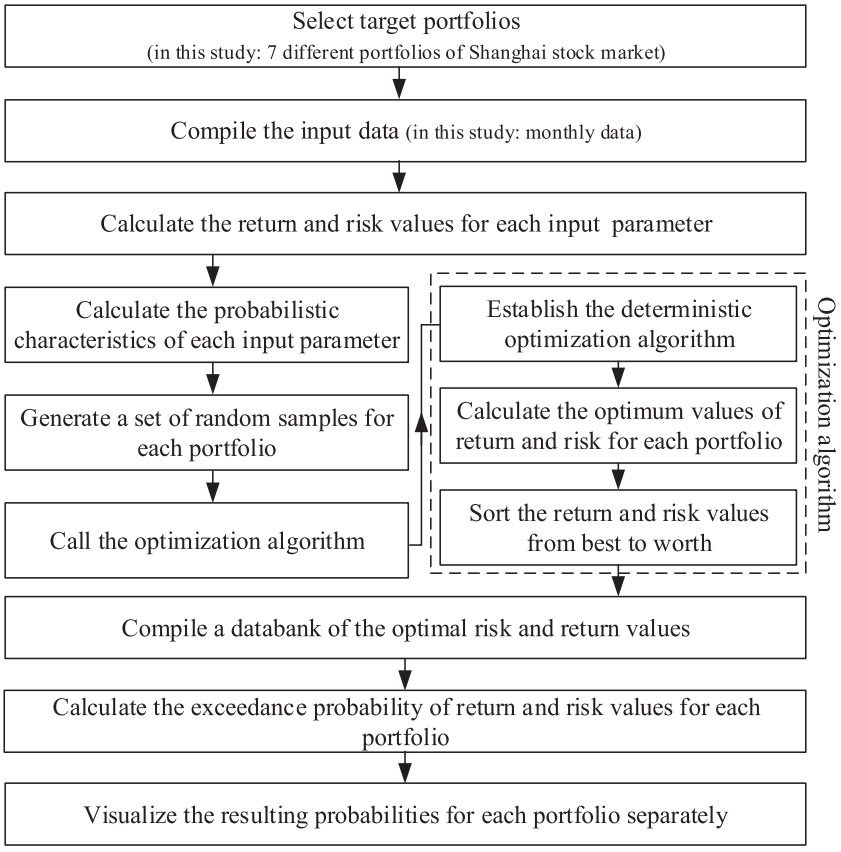



**Fig. 2** A schematic view of the proposed algorithm.

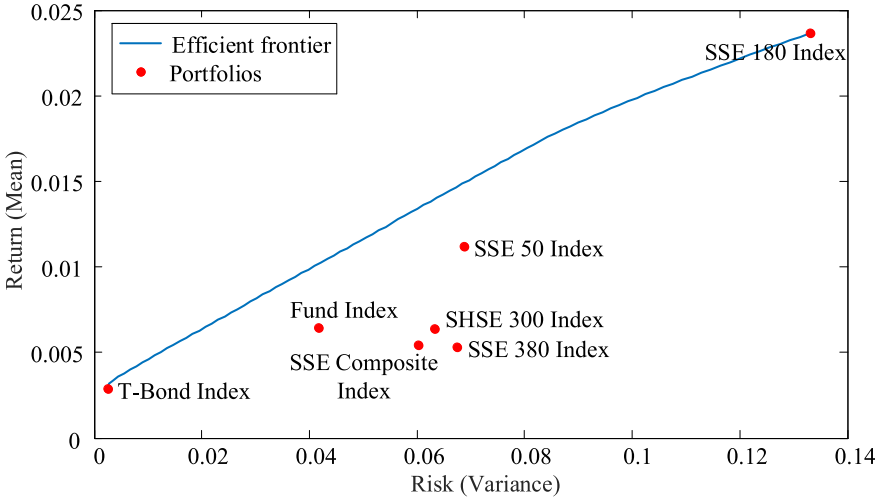



**Fig. 3** Pareto frontier for seven different portfolios from Shanghai stock market.

seven indices where each scenario has a risk and a return. The investors can choose the best combination of stock indices for their investment by specifying the maximum risk or minimum expected return.

### *3.2. Probabilistic modelling of portfolio optimization*

As outlined in the introduction, the Monte Carlo sampling method is adopted in this paper to provide the probabilistic analysis of the established problem. To begin implementing the Monte Carlo algorithm, each of the variables involved in the problem (i.e., the monthly values of each portfolio) should be defined as a random variable. In this way, a relatively wide range of possible scenarios would be covered in the analysis process, and the uncertainties in the model parameters would be imported to the problem formulation. The type of probability distribution function can have a significant effect on the resulting probability. Therefore, the distribution function governing the random variables should be selected according to

the actual data. To this end, the cumulative frequency of daily return data is calculated along with different distribution functions, including Normal, Exponential, Uniform, Logistic, and Gumbel. The results for the return values of portfolio A in 2018/12 is shown in Fig. 4. As can be seen, the Normal distribution provides a closer estimation of original data than other distribution functions. As such, the random variables are modeled by Normal distribution function. The mean and standard deviation values of the variables were calculated from the existing daily data. The results are shown in Figs. 5a and b for the whole 60 months.

Then, following the probabilistic characteristics shown in Fig. 5, a normal random vector was generated for each month separately. Each vector contains 30 components corresponding to daily values. A sample of the generated random vectors for the first portfolio is shown in columns 2 to 10 of Table 2. The summation of the random vectors was then calculated for each month and considered as monthly random values. The results are shown in the last column of Table 2.

By repeating this process, 20,000 random vectors were generated for each portfolio. An overview of 200 random vectors for the first portfolio is given in Fig. 6. Only 200 random sam-

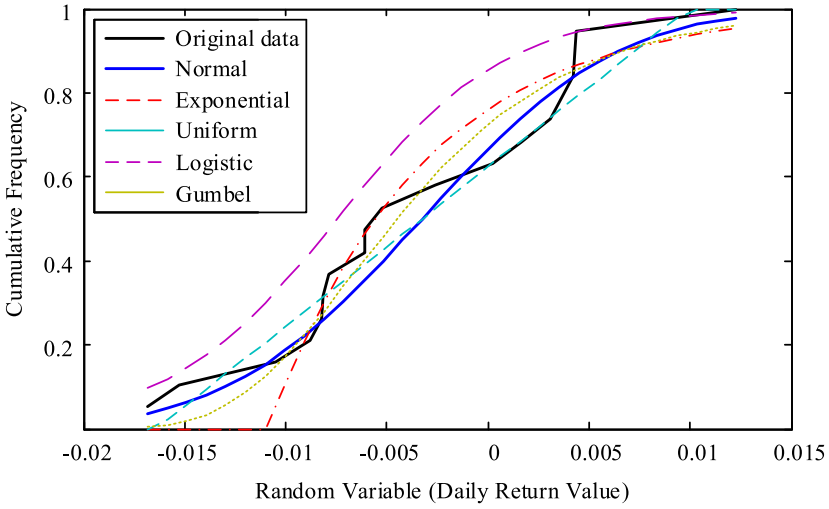



**Fig. 4** Cumulative frequency calculated from the return value of Portfolio A and different distribution functions for the year 2018/12.

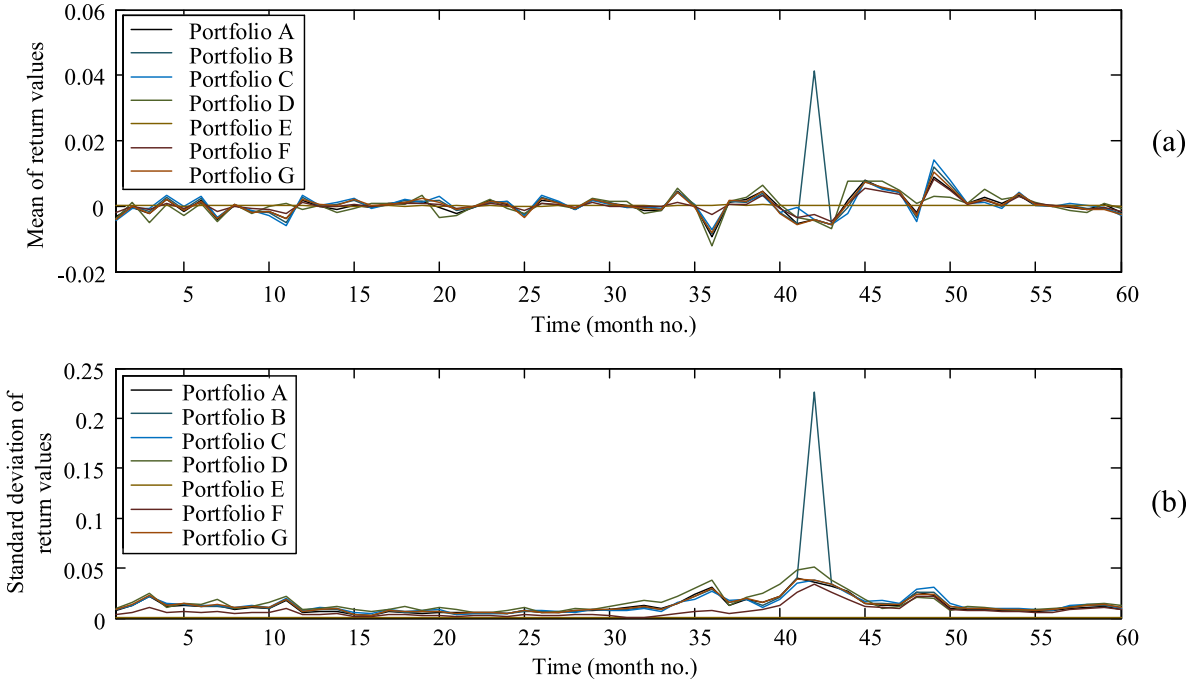



**Fig. 5** Mean and standard deviation of return values for each portfolio in a monthly basis.

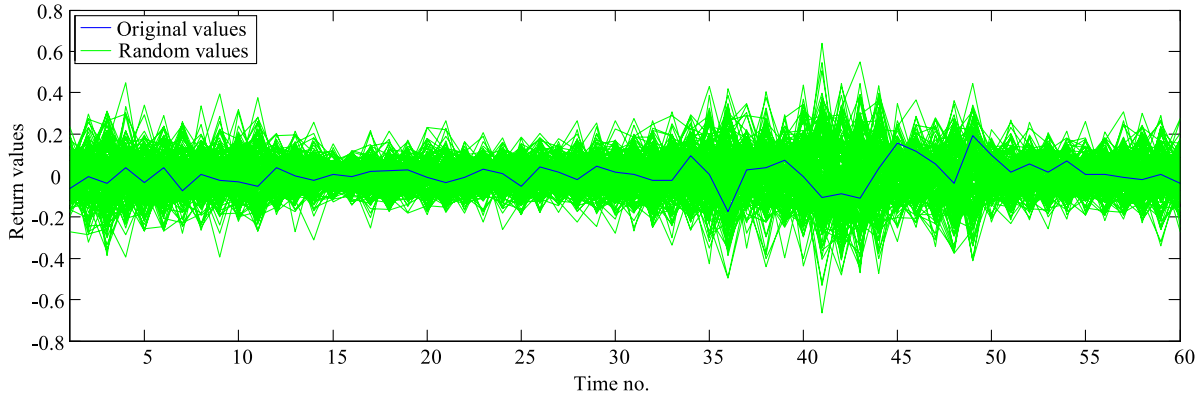



**Fig. 6** Overview of 200 random samples generated for the first portfolio.

ples are used to draw this graph, because the fluctuation of random samples and original data can be seen more clearly.

The random vectors generated for each portfolio were then combined to form a matrix in each random scenario. Subsequently, random matrices were imported into the optimization algorithm, and the corresponding optimal return and risk values were calculated and stored in a vector. A schematic view of this process is shown in Fig. 7.

The histogram of the optimal return and risk values for the first portfolio is shown in Fig. 8. As can be seen, the optimal response of each portfolio is not limited to a certain solution and covers a relatively wide range of possible cases.

To process the output data, the optimal return and risk values obtained from each random scenario should first be sorted from best to worst. The best return value corresponds to a sample with the highest possible value, while the best risk value represents a sample with the lowest value. This means that the return data should be arranged from large to small, while the risk data should be sorted from small to large. For example, the optimal return and risk values obtained from the first random sample are shown in columns 1 to 3 of Table 3. Additionally, the return values (sorted from large to small) are presented in columns 4 and 5 of Table 3 beside the corresponding portfolios. As can be observed, portfolio A ranked 1, portfolio D ranked 7, and the rest of portfolios achieved their own rank. Moreover, the risk values (sorted from small to large) and the corresponding portfolios are shown in columns 6 and 7 of Table 3. In this table, ranks 1 and 7 are assigned to portfolios E and G, respectively, and the rest of the portfolios are located in between. Thus, by sorting the optimal results obtained from the random scenarios, each portfolio received a return and risk rank.

The ranking of both return and risk was then listed for all seven portfolios. The result is shown in a matrix form in Tables 4 and 5.

Then, the number of cases where the rank of return or risk in each portfolio equals $j$ ($j = 1, 2, \ldots, 7$) was counted. Dividing this number by the total number of random scenarios (i.e., 20,000), the probability of reaching rank $j$ for each portfolio is calculated as follows (Eq. 3):

$$P_{ij} = \frac{n_{ij}}{N}, \tag{3}$$

where $n_{ij}$ represents the number of cases in which portfolio $i$ ($i = A, B, \cdots, G$) reaches rank $j$ ($j = 1, 2, \ldots, 7$), and $P_{ij}$ indicates the probability of case $n_{ij}$. The probability for the both risk and return variables is presented in Tables 6 and 7, respectively.

These values were then plotted as a pie chart for each portfolio. The results for the return and risk variables are provided in Fig. 9.

This representation provides a probabilistic statement of ranking for each portfolio that is not limited to a particular

**Table 2** A sample of random vectors generated in each month for the first portfolio.

| Month | Day in each month | | | | | | | | | Summation |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | ⋯ | 27 | 28 | 29 | 30 | |
| 1 | −0.00091 | 0.00207 | −0.01032 | 0.00051 | ⋯ | −0.01728 | −0.01614 | 0.00566 | −0.00271 | −0.11943 |
| 2 | 0.01183 | −0.01057 | 0.00363 | −0.01300 | ⋯ | −0.00395 | −0.00492 | −0.00960 | 0.00683 | −0.01537 |
| 3 | −0.04688 | 0.06419 | 0.02565 | −0.02541 | ⋯ | 0.00847 | −0.03746 | −0.01532 | 0.01916 | −0.03435 |
| 4 | 0.00061 | −0.00721 | −0.00171 | −0.00023 | ⋯ | 0.00856 | 0.00223 | 0.00914 | −0.01156 | 0.07123 |
| 5 | −0.01843 | −0.00989 | 0.01269 | 0.00461 | ⋯ | −0.01433 | −0.00028 | 0.01186 | −0.00496 | −0.04019 |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋯ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| 56 | 0.00069 | 0.01015 | 0.00426 | −0.01124 | ⋯ | −0.00532 | 0.00914 | 0.00002 | −0.00841 | 0.01506 |
| 57 | −0.00352 | 0.01182 | −0.00184 | 0.00425 | ⋯ | 0.00229 | 0.00152 | −0.00769 | −0.00682 | −0.05060 |
| 58 | −0.00546 | 0.00398 | −0.00937 | −0.00764 | ⋯ | −0.00508 | −0.00508 | 0.01082 | 0.00659 | −0.07495 |
| 59 | −0.00542 | −0.00362 | 0.00750 | −0.01254 | ⋯ | 0.00531 | 0.01131 | −0.01767 | 0.00857 | 0.00897 |
| 60 | −0.00721 | −0.01310 | 0.00495 | 0.00201 | ⋯ | −0.01295 | −0.00813 | −0.00475 | −0.01634 | −0.05202 |

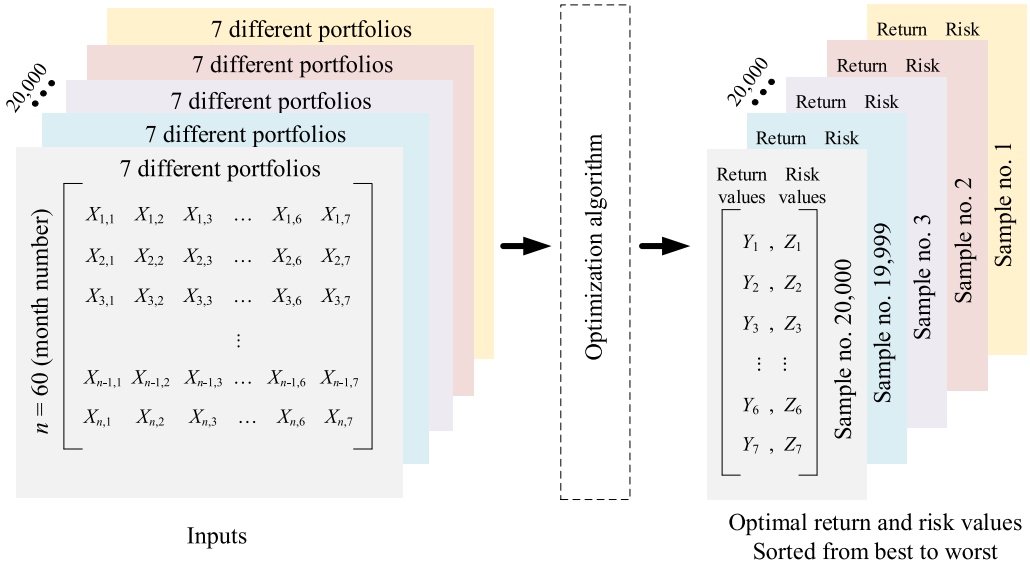



**Fig. 7** Schematic view of input random samples and corresponding optimal values.

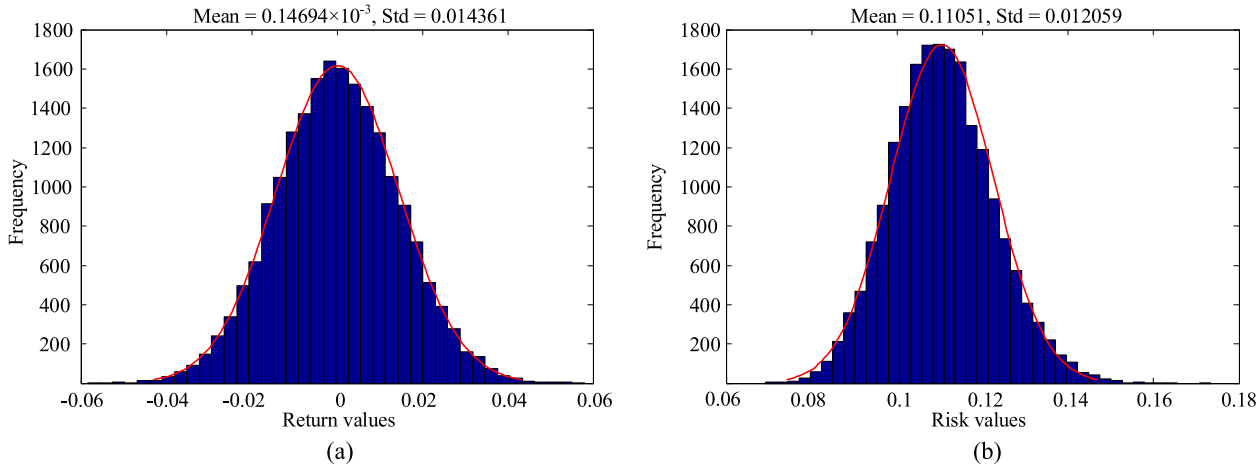



**Fig. 8** Histograms of optimal return and risk values for first portfolio.

**Table 3** Optimal return and risk values obtained from the first random scenario before and after the sorting process.

| Results of optimization algorithm | | | Sorted return | | Sorted risk | |
|---|---|---|---|---|---|---|
| Portfolio | Return | Risk | Portfolio | Return | Portfolio | Risk |
| A | 0.0136 | 0.1168 | A | 0.0136 | E | 0.0168 |
| B | −0.0083 | 0.1107 | G | 0.0118 | F | 0.0893 |
| C | −0.0070 | 0.1187 | E | −0.0013 | D | 0.1029 |
| D | −0.0120 | 0.1029 | F | −0.0059 | B | 0.1107 |
| E | −0.0013 | 0.0168 | C | −0.0070 | A | 0.1168 |
| F | −0.0059 | 0.0893 | B | −0.0083 | C | 0.1187 |
| G | 0.0118 | 0.1265 | D | −0.0120 | G | 0.1265 |

case, but covers a wide range of possible scenarios by taking the uncertainties into account in the both return and risk variables. For example, portfolio B (SSE 180 index) was located in the upper right corner of the deterministic optimization chart (Fig. 3), indicating the best return and the worst risk scenario. The probabilistic chart of this portfolio (Fig. 9) also indicates that this portfolio is very likely to achieve the best rank and the worst risk. However, more information about this portfolio is

**Table 4** Ranking of return and corresponding portfolio at each random scenario.

| Ranking of return | Random scenario | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | ··· | 19996 | 19997 | 19998 | 19999 | 20000 |
| Rank 1 | A | C | D | A | G | ··· | G | B | F | C | C |
| Rank 2 | G | E | A | C | F | ··· | C | F | E | E | G |
| Rank 3 | E | B | F | B | D | ··· | D | E | G | B | F |
| Rank 4 | F | A | B | G | E | ··· | E | D | B | A | E |
| Rank 5 | C | G | G | F | C | ··· | F | G | C | F | B |
| Rank 6 | B | F | E | E | B | ··· | B | A | D | G | A |
| Rank 7 | D | D | C | D | A | ··· | A | C | A | D | D |

**Table 5** Ranking of risk and corresponding portfolio at each random scenario.

| Ranking of risk | Random scenario | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | ··· | 19996 | 19997 | 19998 | 19999 | 20000 |
| Rank 1 | E | E | E | E | E | ··· | E | E | E | E | E |
| Rank 2 | F | F | F | F | F | ··· | F | F | F | F | F |
| Rank 3 | D | G | G | G | A | ··· | A | A | B | C | A |
| Rank 4 | B | D | D | D | C | ··· | C | C | G | G | C |
| Rank 5 | A | C | A | C | G | ··· | D | G | D | A | B |
| Rank 6 | C | A | C | A | D | ··· | G | D | A | B | G |
| Rank 7 | G | B | B | B | B | ··· | B | B | C | D | D |

**Table 6** Estimated probability of return for seven major stock indices.

| Portfolio | Ranking of return | | | | | | |
|---|---|---|---|---|---|---|---|
| | Rank 1 | Rank 2 | Rank 3 | Rank 4 | Rank 5 | Rank 6 | Rank 7 |
| A | 0.1525 | 0.1513 | 0.1331 | 0.1133 | 0.1305 | 0.1542 | 0.1652 |
| B | 0.1970 | 0.1547 | 0.1176 | 0.1031 | 0.1125 | 0.1404 | 0.1750 |
| C | 0.1715 | 0.1539 | 0.1259 | 0.1130 | 0.1260 | 0.1454 | 0.1645 |
| D | 0.1838 | 0.1472 | 0.1150 | 0.1016 | 0.1199 | 0.1464 | 0.1864 |
| E | 0.0167 | 0.0965 | 0.2257 | 0.2989 | 0.2347 | 0.1064 | 0.0212 |
| F | 0.1121 | 0.1496 | 0.1574 | 0.1519 | 0.1518 | 0.1598 | 0.1176 |
| G | 0.1666 | 0.1471 | 0.1254 | 0.1183 | 0.1248 | 0.1476 | 0.1703 |

**Table 7** Estimated probability of risk for seven major stock indices.

| Portfolio | Ranking of risk | | | | | | |
|---|---|---|---|---|---|---|---|
| | Rank 1 | Rank 2 | Rank 3 | Rank 4 | Rank 5 | Rank 6 | Rank 7 |
| A | 0.0 | 0.0437 | 0.2995 | 0.2443 | 0.1975 | 0.1425 | 0.0727 |
| B | 0.0 | 0.0150 | 0.1281 | 0.1438 | 0.1651 | 0.1988 | 0.3493 |
| C | 0.0 | 0.0226 | 0.1957 | 0.2259 | 0.2267 | 0.1997 | 0.1295 |
| D | 0.0 | 0.0056 | 0.0703 | 0.1220 | 0.1814 | 0.2822 | 0.3386 |
| E | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| F | 0.0 | 0.8831 | 0.0824 | 0.0233 | 0.0077 | 0.0031 | 0.0005 |
| G | 0.0 | 0.0301 | 0.2242 | 0.2408 | 0.2216 | 0.1739 | 0.1095 |

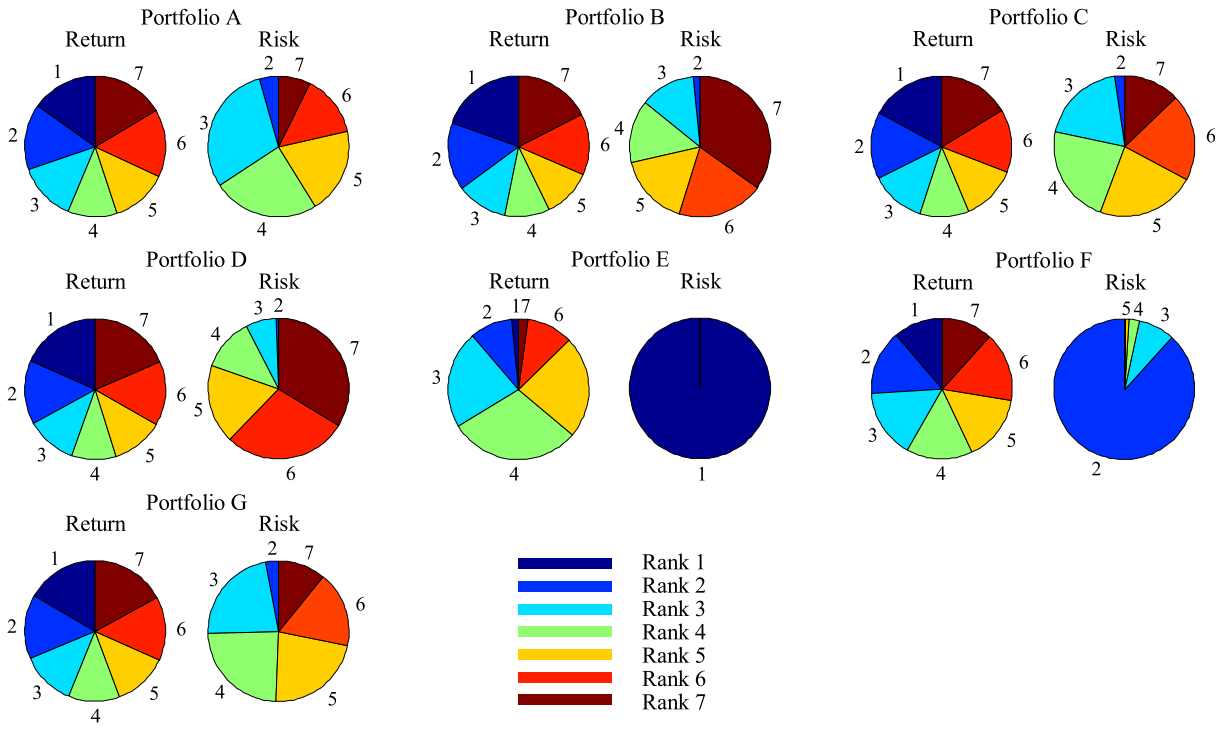



**Fig. 9** Pie chart demonstrating probability of return and risk for seven different portfolios.

**Table 8** The resulting probabilities for the third rank of portfolio A corresponding to different sample sizes.

| Sample size | Return | | Risk | |
|---|---|---|---|---|
| | Number of samples with ranking = 3 | Probability | Number of samples with ranking = 3 | Probability |
| 100 | 15 | 15 | 21 | 21 |
| 500 | 77 | 15.4 | 112 | 22.4 |
| 1000 | 153 | 15.3 | 224 | 22.4 |
| 2000 | 311 | 15.55 | 443 | 22.15 |
| 5000 | 760 | 15.2 | 1118 | 22.36 |
| 10000 | 1480 | 14.8 | 2303 | 23.03 |
| 15000 | 2203 | 14.69 | 3451 | 23.01 |
| 20000 | 2916 | 14.58 | 4582 | 22.91 |

provided in the probabilistic chart, and other probability ranks are also reported. For instance, although the probability of achieving the best return for this portfolio is relatively high ($p = 19.7\%$), the probability of achieving the worst return is also high ($p = 34.93\%$). Such information is also available for all other portfolio ranks.

## 4. Discussion

### 4.1. Convergence assessment

One of the most important points to note is to ensure that the number of random samples used in the Monte Carlo method was sufficient enough to achieve an accurate result. In other words, it should be proved that the number of random samples used in the analysis was sufficient to make the probability converge, so that by increasing the number of random samples, the probability would not change significantly [42]. To this end, the algorithm was repeated for different random sample sizes to evaluate the probability convergence process. The results for the third rank of portfolio A corresponding to different values of random sample sizes are shown in Table 8.

By taking a step smaller than that in Table 8 (i.e., the difference between each two sample size), the resulting probability values were plotted against the number of random samples. The result is shown in Fig. 10. As can be seen, in small random sample sizes, there is a large fluctuation in the probability values. However, with the increase in the size of random samples, the fluctuations decrease. Finally, after using about 12000 random samples, the probability converges, and the fluctuations continue to be very low and negligible.

### 4.2. Robustness of algorithm

In addition to the importance of the accuracy and efficiency of the proposed method, it should also be ensured that if an error

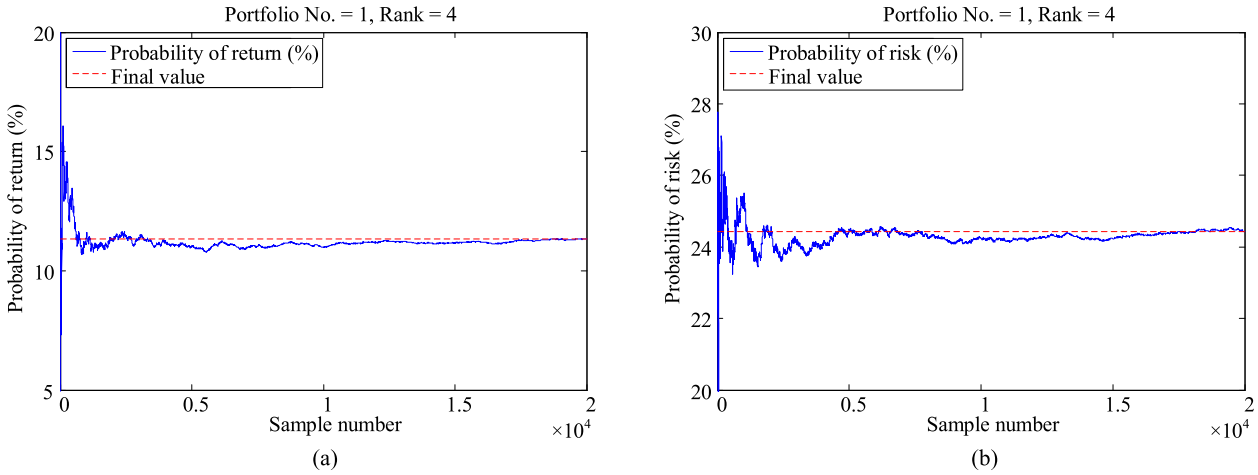



**Fig. 10** Sample number versus probability.

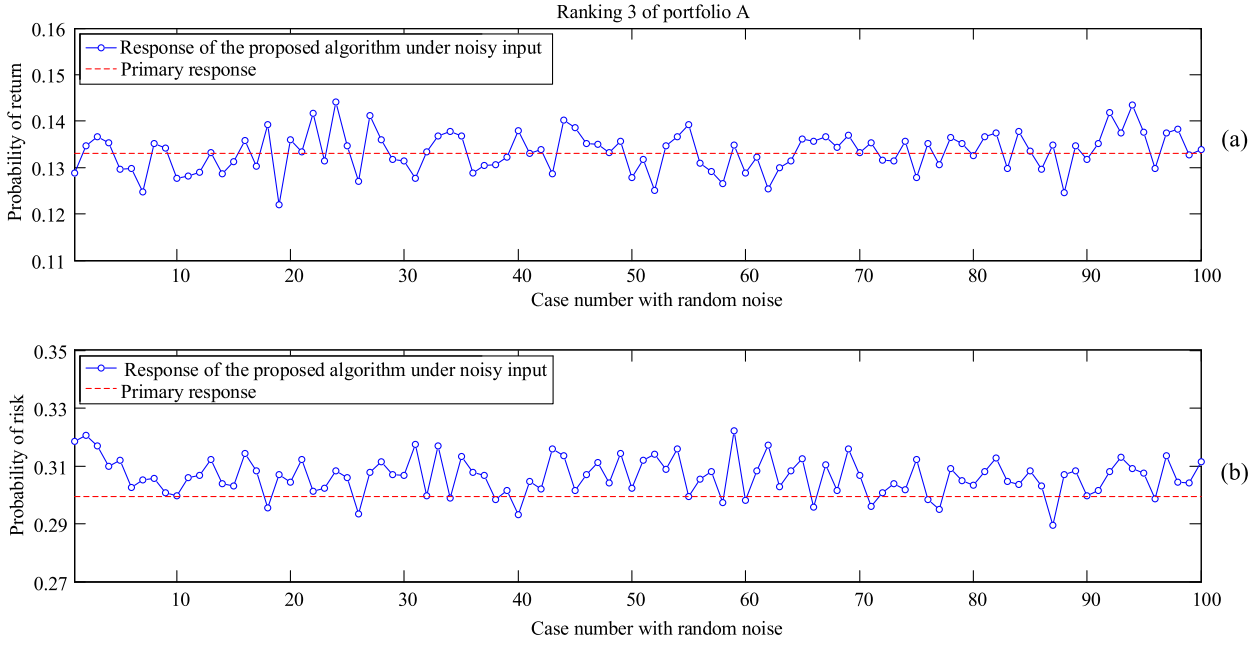



**Fig. 11** Response of proposed algorithm to noisy input, reported for rank 3 of portfolio A: (a) return and (b) risk.

occurs in the model inputs for unintended reasons, the answer will not be affected, and the proposed method will still be able to provide an acceptable answer. This indicates the concept of model robustness [43]. This is very important, because when determining the probabilistic characteristics of variables and distribution functions, the inaccurate estimation of uncertainty can easily lead to a margin of error in the system response, resulting in the unreliable solutions. To numerically investigate this issue, a perturbation function was used to generate a random noise and incorporate it into the variables to evaluate the system response. For this purpose, a Gaussian distribution function with a coefficient of variation of 0.01 was adopted. The mean of the Gaussian function was considered a factor of the mean of random variables (Eq. (4)):

$$\mu_{f_i} = \alpha \times \mu_i, \tag{4}$$

where $\mu_i$ represents the mean of random variable $i$, $\alpha$ denotes the noise contribution factor, and $\mu_{f_i}$ is regarded as the mean of the Gaussian distribution function for variable $i$. Assuming $\alpha = 2\%$, a series of 100 random noises were generated and added to the initial data. The proposed algorithm was then utilized to calculate the optimal probabilistic solution for all seven portfolios under the noise-polluted inputs. The results for the rank 3 of return and risk in portfolio A and the histogram of the absolute error are shown in Figs. 11 and 12, respectively. Additionally, the mean and standard deviation of both probability and the absolute error are calculated and reported in Table 9.

As can be seen, the proposed algorithm exhibits a very robust response against the applied changes, so that the obtained results are very close to the actual values in almost all cases, and the resulting error is close to zero. This is because

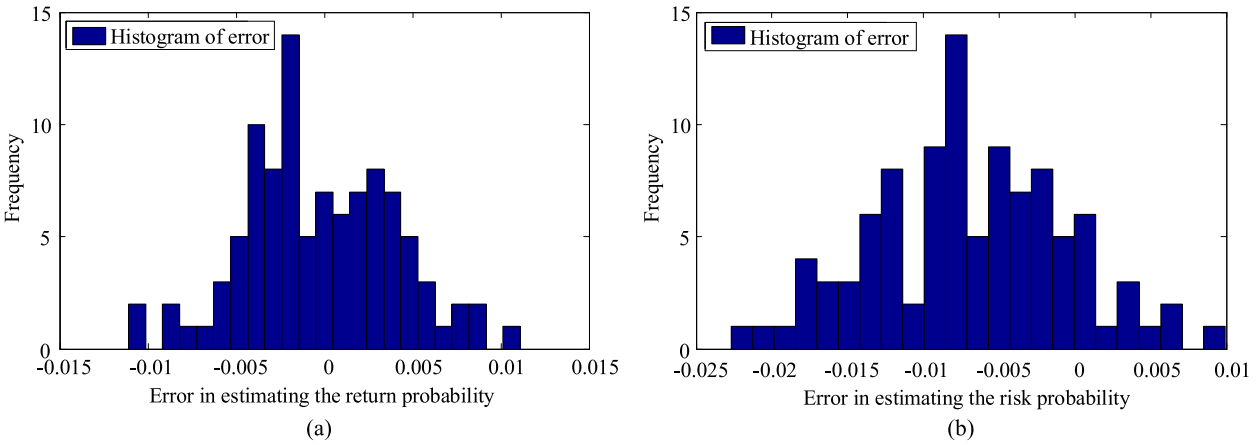



**Fig. 12** Histogram of error in estimating the probability of (a) return and (b) risk.

**Table 9** Mean and standard deviation of both probability ($P$) and absolute error ($\Delta P$).

| Target parameter | Mean of $P$ | Standard deviation of $P$ | Mean of $\Delta P$ | Standard deviation of $\Delta P$ |
|---|---|---|---|---|
| Return | 0.13347 | 0.00431 | 0.00037 | 0.00431 |
| Risk | 0.30678 | 0.00646 | 0.00728 | 0.00646 |

the noise applied to the input variables is accommodated by the algorithm, and consequently, no significant change is made in the results.

## 5. Conclusion

In this paper, a probabilistic form of portfolio optimization was provided for Shanghai stock market considering the uncertainty in the return and risk values. For this purpose, a comprehensive database of input values was collected on a daily basis from January 2014 to December 2018. Then, the mean and standard deviation of the data were calculated for each month and imported to the Monte Carlo algorithm as a vector of random variables. Afterwards, by generating a vector of random numbers for each random variable, 20,000 random scenarios were defined, and the optimal results were separately calculated for each one. Then, by sorting the optimal return and risk values, the rank of each portfolio and the corresponding probabilities were calculated. The results were then presented as a set of two probabilistic pie charts for the return and risk ranks. The most important contributions of this paper can be summarized as follows:

- Instead of direct use of the dataset collected from the stock market, the monthly values corresponding to each portfolio were defined as random variables with the specified means and standard deviations to incorporate the uncertainty into the optimization process.
- By introducing the optimization algorithm into the Monte Carlo sampling method, a systematic approach was presented to compile a database of optimal returns and risks.
- By sorting the obtained artificial data, a simple definition of optimal ranking was presented to calculate the corresponding probability for all seven portfolios.
- By applying a Gaussian random noise to the input data and evaluating the fluctuation of the results, the robustness of the proposed algorithm was investigated.

The results of this paper are essential for evaluating the desired stock indices and the amount of investment in each portfolio, providing a straightforward criterion for the decision-makers and investors. Finally, it should be noted that one of the simplified assumptions in this paper is that the input portfolios are assumed to be independent. In case of having correlations between the inputs, it should be seen when using the Monte Carlo algorithm and generating random samples. If the correlation is linear, it is sufficient to use the Pearson correlation coefficient and the covariance matrix to model the correlation. However, if the correlation is nonlinear, other methods such as Copula functions or joint distribution function governing random input variables can be used to generate random samples. The probabilistic portfolio optimization assuming the correlations between different portfolios is an interesting and challenging topic to be considered by the authors in the future.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

**Appendix A.** The seven portfolio indices used in the paper are as follows. More information about indices can be found in [44].

**SSE Composite Index 000001:** SSE composite index consists of all the stocks (including A-shares and B-shares)

listed on SSE. The index aims to reflect the overall performance of the Shanghai stock market. The index is weighted by total market capitalization to reflect the price performance of listed stocks in SSE. The index was launched on July 15, 1991.

**SSE 180 Index 000010:** SSE 180 index consists of the 180 largest and most liquid A-share stocks listed on SSE. The index aims to reflect the performance of the Shanghai blue chips. SSE 180 indices are composed of the constituents with good performance, without serious financial problems or large price volatility that shows strong evidence of manipulation.

**SSE 50 Index 000016:** SSE 50 index consists of the 50 largest and most liquid A-share stocks listed on SSE. The index aims to reflect the overall performance of the most influential leading Shanghai stocks. SSE 50 indices are selected from SSE 180 Index constituents and composed of 50 Shanghai stocks with large market cap and good liquidity.

**SSE 380 Index 000009:** SSE 380 index consists of 380 stocks with mid-size market cap, high growth and good profitability which are selected from the remaining Shanghai listed A-shares after deleting the constituents of SSE 180 index. The index aims to comprehensively reflect the performance of the new Shanghai blue chips. SSE 380 indices are composed of Shanghai stocks from the SSE 180 index universe except for certain stocks. The constituents are selected by ranking of revenue growth, return on net assets, turnover, and total market value.

**SSE Fund Index 000011:** The constituents for SSE fund index are all the securities investment funds listed on SSE. The base day for SSE fund index is May 8, 2000. The base period is the total market capitalization of all securities investment funds of that day. The base value is 1000. The index was launched on June 9, 2000.

**T-Bond index:** Treasury bonds, a kind of government bonds issued for raising financial funds, are the certificates for the claims and debts issued to investors by the central government. In the exchange market, there are mainly two kinds of treasury bonds: coupon-bearing treasury bonds and discount treasury bonds.

**SHSE-SZSE300 Index:** SZSE 300 index and its sub-indices (SZSE 100 Price Index and SZSE 200 Price Index) are the stock market indices of Shenzhen Stock Exchange, representing top 300 companies by free-float adjusted market capitalization. The sub-indices represent the top 100 companies and next 200 (the 101st to 300th) companies, respectively. SZSE 300 index itself is a sub-index of SZSE Component Index, SZSE 1000 Index and SZSE Composite Index.

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